

Fa3

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```
supplied_images <- c(0.15, 0.20, 0.25, 0.40)
relevant_images <- c(0.50, 0.60, 0.80, 0.85)
probability <- supplied_images * relevant_images
probability
```

```
## [1] 0.075 0.120 0.200 0.340
```

```
overall_relevant <- sum(probability)
overall_relevant * 100
```

```
## [1] 73.5
```

```
#Tossing a coin twice
sample_space <- c("HH", "HT", "TH", "TT")
#Events
E1 <- c("HH", "TT")
E2 <- c("HH", "HT")
E3 <- c("TH", "HH")
#individual probabilities
p_E1 <- length(E1) / length(sample_space)
p_E2 <- length(E2) / length(sample_space)
p_E3 <- length(E3) / length(sample_space)
#overlapping outcomes
E1_and_E2 <- intersect(E1, E2)
E2_and_E3 <- intersect(E2, E3)
E1_and_E3 <- intersect(E1, E3)
#combine
p_E1_and_E2 <- length(E1_and_E2) / length(sample_space)
p_E2_and_E3 <- length(E2_and_E3) / length(sample_space)
p_E1_and_E3 <- length(E1_and_E3) / length(sample_space)
#pairwise independence
p_E1_and_E2 == (p_E1 * p_E2)
```

```
## [1] TRUE
```

```
p_E2_and_E3 == (p_E2 * p_E3)
```

```
## [1] TRUE
```

```
p_E1_and_E3 == (p_E1 * p_E3)
```

```
## [1] TRUE
```

```
#mutual independece
```

```
E1_and_E2_and_E3 <- intersect(E1_and_E2, E3)
```

```
p_all_three <- length(E1_and_E2_and_E3) / length(sample_space)
```

```
p_all_three == (p_E1 * p_E2 * p_E3)
```

```
## [1] FALSE
```